

## Chapter 4

### Worksheet

#### Instructions:

Please complete all activities in this worksheet. Write your answers in the spaces provided or on separate sheets if needed.

#### Worksheet for Chapter 4: Basics of Data Analysis

**Instructions:** Complete each section of this worksheet individually or in small groups as directed by your teacher. Use a calculator or digital tool (like Google Sheets) if needed for calculations. Be prepared to discuss your answers in class or present your findings. Have fun exploring data like a true detective!

#### Part 1: Spotting Patterns and Outliers (15 Points)

**Objective:** Practice identifying patterns and outliers in real-world data scenarios.

- **Track and Analyze (10 Points)** You've tracked the number of hours you spent on social media over 7 days:
  - Day 1: 2 hours
  - Day 2: 1.5 hours
  - Day 3: 2 hours
  - Day 4: 1.8 hours
  - Day 5: 2.2 hours
  - Day 6: 5 hours
  - Day 7: 1.9 hours
- **a. Identify the Pattern (5 Points):** Describe any pattern you see in the data. Is there a trend (increasing, decreasing, or cyclical)? Write 2-3 sentences.
- **b. Spot the Outlier (5 Points):** Which data point is an outlier? Why do you think it stands out, and what might explain it? Write 2-3 sentences.
- **Quick Think (5 Points)** Why is it important to investigate outliers instead of ignoring them? Give one real-life example (e.g., in sports, school, or business).

#### Part 2: Crunching Numbers with Stats (20 Points)

**Objective:** Calculate and interpret basic statistical measures.

- **Calculate and Interpret (15 Points)** Your group of friends shared how many minutes they spend gaming daily:
  - Friend 1: 30 min
  - Friend 2: 45 min
  - Friend 3: 30 min
  - Friend 4: 60 min

- Friend 5: 120 min
- Friend 6: 45 min

- a. **Mean (5 Points):** Calculate the average gaming time. Show your work.

Answer: \_\_\_\_\_ minutes

- b. **Median (5 Points):** Find the middle value after sorting the data. Show your sorted list. Sorted List: \_\_\_\_\_ Answer: \_\_\_\_\_ minutes
- c. **Mode (3 Points):** Identify the most common value. Answer: \_\_\_\_\_ minutes
- d. **Range (2 Points):** Calculate the difference between the highest and lowest values. Answer: \_\_\_\_\_ minutes
- **Interpret the Stats (5 Points)** Based on your calculations, which measure (mean, median, or mode) best describes the "typical" gaming time for most friends? Why? Write 2-3 sentences. (Hint: Consider the effect of outliers.)

### Part 3: Making Sense of Visuals (15 Points)

**Objective:** Draw conclusions from data visualizations and avoid interpretation traps.

- **Analyze a Chart (10 Points)** Imagine a bar chart showing the number of students in your class who prefer different snacks:
  - Chips: 15 students
  - Candy: 8 students
  - Fruit: 5 students
  - Popcorn: 2 students
- a. **Conclusion (5 Points):** What conclusion can you draw from this chart? Suggest one action your school canteen could take based on this data. Write 2 sentences.
- b. **Possible Trap (5 Points):** What is one interpretation mistake to avoid when looking at this data? (Hint: Think about sample size, context, or assumptions.) Write 2 sentences.
- **Spot the Mistake (5 Points)** A line graph shows your city's temperature rising from 20°C to 25°C over a week, and a friend says, "This proves global warming is happening right now!" Why might this conclusion be a mistake? Write 2-3 sentences explaining the error.

### Part 4: Creative Application (10 Points)

**Objective:** Apply data analysis skills to a personal or fun scenario.

- **Mini Data Project (10 Points)** Pick something you can track over 3-5 days (e.g., hours of sleep, number of steps walked, or snacks eaten). Write down your plan below:
  - What will you track? \_\_\_\_\_
  - How will you record it (notebook, app)? \_\_\_\_\_
  - Predict one pattern or outlier you might find. \_\_\_\_\_

After tracking, calculate at least two stats (like mean and range) and write one conclusion. (If time is limited, imagine fake data and analyze it.)

- Stat 1: \_\_\_\_\_
- Stat 2: \_\_\_\_\_
- Conclusion: \_\_\_\_\_

### Part 5: Reflection (5 Points)

**Objective:** Think critically about what you've learned.

- **Think Deep (5 Points)** How can data analysis help you in everyday life (like with school, hobbies, or decisions)? Write 2-3 sentences with a specific example.

**Bonus Challenge (Optional, +5 Points):** Create a simple hand-drawn or digital chart based on your mini data project in Part 4. Label it clearly and write one sentence explaining what it shows. Attach it or describe it here.